

12. Newman HTML Report Generation:

In API automation testing, executing tests is not enough — we also need **proper reports** to analyze results. In this project using Fake Store API, we use **Newman HTML reports** to generate a clear and professional test execution summary.

12.1 What is Newman HTML Report?

A Newman HTML report is a **visual test execution report** generated after running a Postman collection using Newman CLI.

☞ It converts raw CLI output into:

- Structured HTML dashboard
 - Graphical summary
 - Pass/fail status
 - Request-wise execution details
-

12.2 Why HTML Reports are important?

HTML reports are used because:

- Easy to understand for non-technical stakeholders
 - Provides visual summary of test execution
 - Helps in debugging failures quickly
 - Useful for QA documentation
 - Can be shared with clients and managers
-

12.3 Tools used for reporting

We mainly use:

◆ Newman CLI

Runs the collection

◆ HTML Reporter Plugin

Example:

- newman-reporter-html
 - newman-reporter-htmlextra (advanced)
-

12.4 Installation of HTML Reporter

Before generating reports, install reporter:

```
npm install -g newman-reporter-htmlextra
```

🔗 This installs advanced reporting plugin.

12.5 Command to Generate HTML Report

◆ Basic command:

```
newman run collection.json -e environment.json -r htmlextra
```

◆ Output location example:

After execution, report is generated like:

```
/newman/report.html
```

12.6 What HTML Report Contains

The report includes:

◆ 1. Execution Summary

- Total requests
 - Passed tests
 - Failed tests
 - Total execution time
-

◆ 2. Request-wise Details

Each API shows:

- Request URL
 - Method (GET, POST, PUT, DELETE)
 - Status code
 - Response time
-

◆ 3. Test Results

- Assertions passed/failed
 - Error messages if any test fails
-

◆ 4. Environment Details

- Base URL used
 - Token values
 - Variable data
-

◆ 5. Graphical Charts

- Pass vs Fail graph
 - Time distribution chart
 - Request performance chart
-

12.7 Example Flow in This Project

When we run Newman report for Fake Store API:

Step 1: Authentication API

- Login request executed
 - Token generated
-

Step 2: Product APIs

- Get all products → PASS

- Add product → PASS
 - Update product → PASS
-

Step 3: Cart APIs

- Add to cart → PASS
 - Delete cart → PASS
-

Step 4: Final Report Generated

- All API results shown in HTML format
-

12.8 Advantages of HTML Reports

- Professional reporting format
 - Easy to share in teams
 - Improves QA documentation
 - Helps in client reporting
 - Useful for audit and tracking
 - Better than CLI logs
-

12.9 Limitations

- Requires extra plugin installation
 - Large reports for big collections
 - Slight performance overhead
 - Not real-time (generated after execution)
-

12.10 Real-Time Use in Industry

In real companies:

- QA runs Newman in CI/CD pipeline
- HTML reports are sent to:
 - Email
 - Slack
 - Jenkins dashboard

- Used for daily regression reports
-

✓ Final Interview Summary

In this project, I used Newman HTML Extra reporter to generate detailed execution reports for the Fake Store API test suite. After running Postman collections via Newman CLI, I generated HTML reports containing request details, status codes, response times, test results, and graphical summaries. These reports helped in better visualization of API test execution, improved stakeholder communication, and supported CI/CD reporting in automation pipelines.
